

BFCL v3 Multi-Turn Benchmark Report

AgentOpt Model Selection Framework

9 Bedrock Models | 200 Samples | LangGraph Agent | March 2026

1. Brute Force Ground Truth (LangGraph)

- Exhaustive evaluation: all 9 models x 200 BFCL v3 multi-turn samples.
- Single-agent architecture: one LLM handles multi-turn function calling.
- Best model: Claude Opus 4.6 at 72.00% accuracy.
- Total brute force cost across all models: \$85.42.
- Total evaluations: 1,800 (9 models x 200 samples).

Rank	Model	Accuracy	Avg Lat	Total Cost	Cost/Sample
1	Claude Opus 4.6	72.00%	223s	\$60.78	\$0.3039
2	Qwen3 Next 80B A3B	71.00%	226s	\$1.87	\$0.0094
3	Kimi K2.5	68.50%	228s	\$3.86	\$0.0193
4	Claude Haiku 4.5	65.00%	209s	\$11.97	\$0.0598
5	gpt-oss-120b	61.00%	209s	\$1.13	\$0.0056
6	Qwen3 32B	50.00%	211s	\$0.97	\$0.0049
7	Claude 3 Haiku	45.00%	205s	\$3.42	\$0.0171
8	gpt-oss-20b	39.00%	204s	\$0.44	\$0.0022
9	Minstral 3 8B	33.50%	214s	\$0.98	\$0.0049

2. Official BFCL Leaderboard Comparison

- Our scores differ from the official BFCL v3 leaderboard due to eval methodology.
- Three bugs were found and fixed in our v5 eval (see Section 5 for details).
- Remaining gap is due to live vs recorded execution (architectural difference).
- Our agentic eval rewards adaptive tool use (models see real return values, can retry and adapt). The official eval rewards structured output accuracy (models never see real feedback). This benefits Opus (+9% over our score), which excels at structured generation.
- Live execution introduces state divergence from retries and side effects that the official eval would never see (both model and GT are re-executed on fresh instances).
- Note: Qwen3 Next 80B A3B is not on the official leaderboard. The nearest entry (Qwen3-30B-A3B-Instruct-2507, 43.5%) is a different, smaller model.
- The ranking differences across eval methodologies illustrate that model selection depends on the evaluation environment -- a core motivation for AgentOpt.

Model	Our Score	Official	Delta	Notes
Claude Opus 4.6	72.0%	81.0%	-9.0%	
Qwen3 Next 80B A3B	71.0%	43.5%	+27.5%	Not on leaderboard; nearest is 30B-A3B-2507
Kimi K2.5	68.5%	62.0%	+6.5%	Official uses K2 (different version)
Claude Haiku 4.5	65.0%	63.5%	+1.5%	

Qwen3 32B	50.0%	54.0%	-4.0%	Both use prompting mode
Ministral 3 8B	33.5%	0.0%	+33.5%	We beat official 0%

gpt-oss-20b, gpt-oss-120b, Claude 3 Haiku: not on official leaderboard (Amazon-specific or delisted).

3. Offline Selector Simulation (50 Seeds)

- Per-sample scores from brute force replayed with each selector's logic
 - zero additional API calls required.
- Each selector simulated 50 times with different random seeds.
- Hill climbing uses topology-aware neighbors (quality/speed ranking).
 - HC (1) = 1 random restart; HC (3) = 3 restarts (library default).
- Metrics: Find Rate = % finding true best. Mean Acc = avg accuracy of chosen model.

Selector	Found Best	Mean Acc	Evaluations	Cost	Savings
Brute Force	100%	72.00%	1,800	\$85.42	--
Arm Elimination	100%	72.00%	922	\$76.33	11%
Hill Climbing (3)	94%	71.94%	1,652	\$80.38	6%
Epsilon LUCB	60%	71.33%	407	\$42.60	50%
Hill Climbing (1)	24%	70.89%	808	\$27.49	68%
Bayesian Opt.	56%	70.61%	1,000	\$50.98	40%
Random Search	36%	67.99%	600	\$31.62	63%
Threshold SE	12%	57.52%	285	\$6.45	92%
LM Proposal	0%	45.00%	200	\$3.42	96%

4. Key Takeaways

Brute Force Insights

- Opus and Qwen3 Next are statistically tied (72% vs 71%, ~2 samples difference).
 - Qwen3 Next achieves near-Opus accuracy at 32x lower cost (\$1.87 vs \$60.78).
- Top 4 tightly clustered (65-72%) -- BFCL multi-turn is noisy and non-deterministic.
- Clear quality tiers: Opus/Qwen3 Next/Kimi (68-72%) > Haiku 4.5/gpt-oss-120b (61-65%)
 - > Qwen3 32B/C3 Haiku (45-50%) > gpt-oss-20b/Ministral (34-39%).
- Opus dominates cost: \$60.78 of \$85.42 total (71%). Drives selector behavior.

Selector Insights

- Arm Elimination is again perfect (100% find rate), consistent across all benchmarks.
- Cost savings are poor across the board (Arm Elim: 11%, HC(3): 6%).
 - This is because the best model (Opus) is also the most expensive.
 - Selectors cannot avoid evaluating Opus, and evaluating Opus alone eats ~71% of budget.
 - Contrast with HotpotQA: best combo (Ministral+Opus) is cheap on the planner side, so eliminating expensive planner combos early saves real money (64% savings).
- Selector efficiency depends on whether the best model is also the most expensive.
 - When it is (BFCL), there is no shortcut. When it is not (HotpotQA), savings are large.
- Epsilon LUCB is the best budget option: 60% find rate with 50% savings.
- LM Proposal fails completely (0% find rate): GPT-4.1 picked Claude 3 Haiku (rank 7 of 9, 45%) every seed. Cannot predict multi-turn FC performance from task previews alone -- base knowledge of model capabilities is insufficient.

5. Eval Methodology: Why Our Scores Differ from Official BFCL

Live Execution vs Recorded Re-Execution

- Official BFCL records model tool calls as strings during inference.
At eval time, both model and GT call strings are re-executed on fresh backends.
The model never sees real return values or error messages during the conversation.
- Our eval uses LangGraph, which executes tool calls live against backend instances.
The model sees real return values, can retry failed calls, and adapts its strategy.
This is more realistic (real agents call tools live) but creates two divergences:
(1) Retries and error recovery leave side effects that cause state mismatches with GT.
(2) Models can succeed through retry paths the GT does not anticipate.
- Our eval measures agentic tool use; official BFCL measures structured output generation.
Neither is wrong -- they test different capabilities.

Bugs Found and Fixed (v4 -> v5)

- Bug 1 (Critical): `book_flight()` parameter mismatch.
The `func_doc` declared `travel_cost` as required, but the backend did not accept it.
ALL models scored 0/41 on samples using `book_flight` (20.5% penalty).
In official eval, both model and GT calls fail identically (re-execution), so states match. In our live eval, smart models retry without the bad param and succeed, creating a state mismatch. Fixed by adding `travel_cost` as an accepted parameter.
- Bug 2 (Major): Missing 'continue calling tools' system prompt instruction.
Official BFCL tells models: 'Continue to output functions to call until you have fulfilled the user's request.' Our prompt lacked this. Failure rate correlated with turn count (20% at 2-3 turns, 50% at 5+ turns). Fixed by adding the instruction.
- Bug 3 (Minor): `delete_message()` parameter mismatch.
Similar to Bug 1 but only affected 5 samples. Fixed by adding `message_id` parameter.

Impact of Fixes

Model	v4 (Buggy)	v5 (Fixed)	Delta
Claude Opus 4.6	56.5%	72.0%	+15.5%
Qwen3 Next 80B A3B	58.0%	71.0%	+13.0%
Kimi K2.5	52.5%	68.5%	+16.0%
Claude Haiku 4.5	53.0%	65.0%	+12.0%
gpt-oss-120b	43.5%	61.0%	+17.5%
Qwen3 32B	42.0%	50.0%	+8.0%
Claude 3 Haiku	37.0%	45.0%	+8.0%
gpt-oss-20b	29.0%	39.0%	+10.0%
Ministral 3 8B	28.5%	33.5%	+5.0%